whereas, active transport uses cellular energy to move them against a gradient. The transport proteins discussed in facilitated diffusion can also transport ions or molecules against a concentration gradient with the expenditure of cellular energy as an active process. Pumps use a source of free energy such as ATP or light to drive the thermodynamically uphill transport of ions or molecules. The pump action is an example of active transport. Example: Na^+-K^+-ATPase pump (Table 11.1).

Table 11.1 Comparison of different transport mechanisms

Property	Passive transport		Active transport
	Simple diffusion	Facilitated diffusion	
Nature of process	Physical	Biological	Biological
Requirement for presence of membrane protein	No	Yes	Yes
Selectivity of molecule	No	Yes	Yes
Saturation of transport	No	Yes	Yes
Uphill transport	No	No	Yes
Energy requirement (ATP)	No	No	Yes
Sensitivity to inhibitors	No	Yes	Yes

Check your grasp!

What are the similarities and differences between co- transport and counter transport?

Solution:

Similarity: In both system two molecules are involved for the unidirectional transport.

Difference: In co-transport, two molecules are transported together whereas, in counter transport two molecules are transported in opposite direction to each other.

11.3 Plant Water Relations

Water plays an essential role in the life of the plant. The availability of water influences the external and internal structures of plants as protoplasm is made of 60-80% water. Water is a **universal solvent** since most of the substances get dissolved in it and the high tensile strength of water molecule is helpful in the ascent of sap. Water maintains the internal temperature of the plant as well as the turgidity of the cell.

11.3.1 Imbibition

Colloidal systems such as **gum, starch, proteins, cellulose, agar, gelatin** when placed in water, will absorb a large volume of water and swell up. These substances are called **imbibants** and the phenomenon is **imbibition**.

Examples: 1. The swelling of dry seeds 2. The swelling of wooden windows, tables, doors due to high humidity during the rainy season.



The Power of Imbibition

In olden days, small wooden pegs were inserted into crevices of rocks followed by continuous hydration. Due to imbibition the volume of wooden peg increases and cuts off rocks precisely.

The gluten from wheat can take as much as 300% of its own weight

Significance of imbibition

i. During germination of seeds, imbibition increases the volume of seed enormously and leads to bursting of the seed coat.

ii. It helps in the absorption of water by roots at the initial level.

Activity

Imbibition experiment

Collect 5 gm of gum from Drumstick tree or Babool tree or Almond tree. Immerse in 100ml of water. After 24 hours observe the changes and discuss the results with your teacher.



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